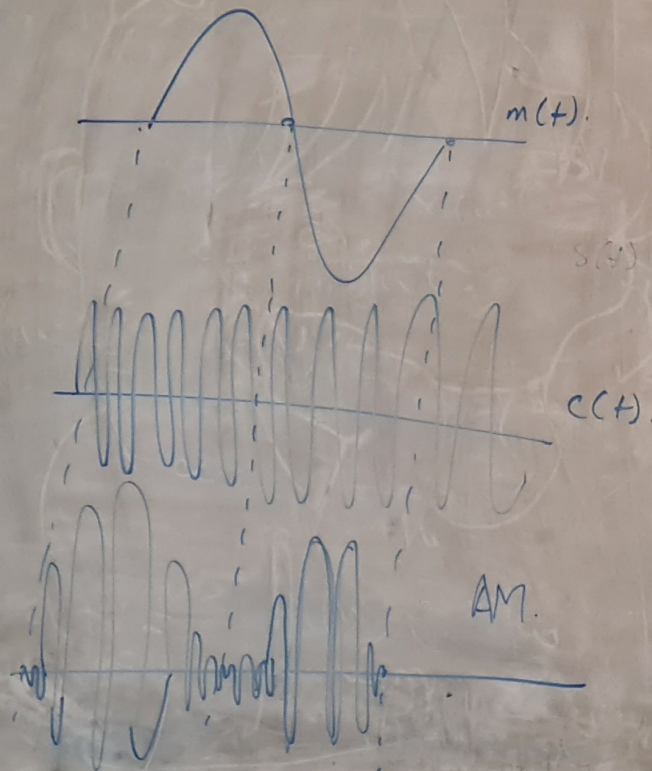


Angle Modulation FM / PM

$$S(t) = A_c \cos(\phi_i)$$

$\phi_i \propto m(t)$



$$S(t) = (A_c \cos \omega_c t) (A_m \cos \omega_m t)$$

In Tx.
voltage control osc.

Convert $(V) \rightarrow (f)$

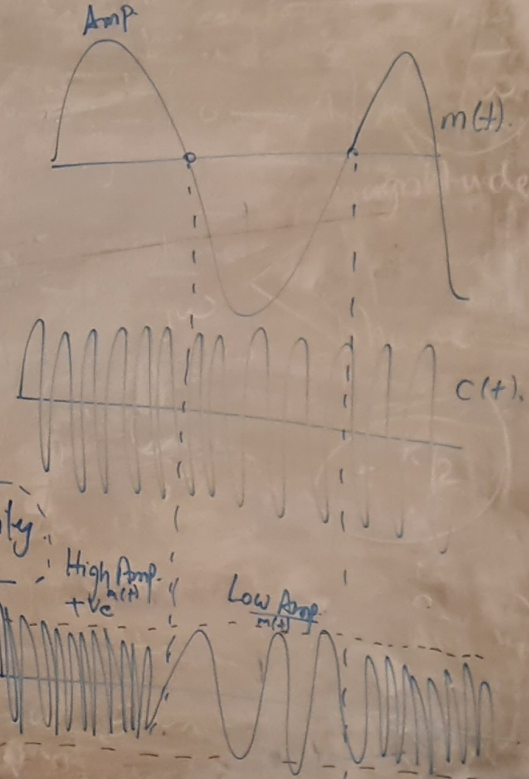
$$f_i = f_c + A_m k_f$$

frequency sensitivity

- at Rx.

Zero crossing detector

+ Freq to Voltage
convert

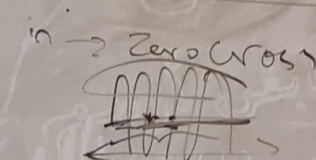
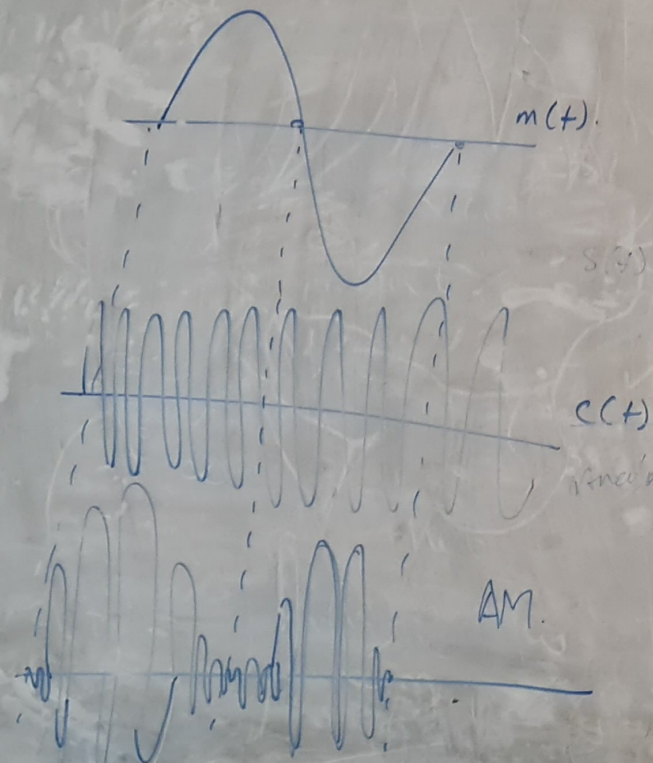


Angle Modulation

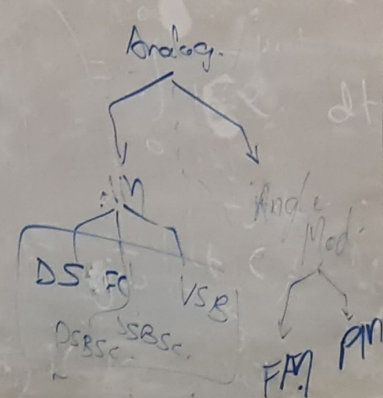
FIM / PM

$$S(t) = A_c \cos(\omega_c t + \phi(t))$$

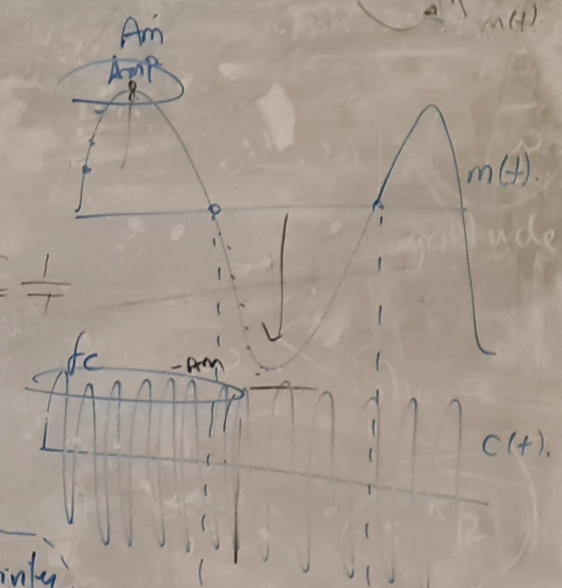
$\phi(t) \propto \int m(t) dt$



$$S(t) = \underbrace{(A_c \cos \omega_c t)}_{\text{Carrier}} \underbrace{(A_m \cos \omega_m t)}_{\text{Message}}$$

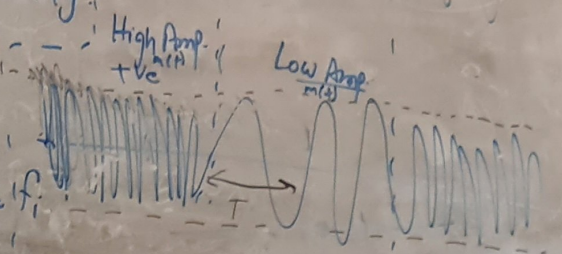


in Tx.
voltage control osc. ω_c
Convert $(V) \rightarrow (f)$
 $f_i = f_c + A_m(k_f)$
instantaneous freq
- at Rx.



Zero crossing detectors.

+ Freq to Voltage convert f_i



Angle Modulation

FIM / PM

$$S(t) = H_c \cos(\phi(t))$$

$\phi(t) = 2\pi \int m(t) dt$

In Tx.
voltage control osc. (VCO)

$$f = \frac{1}{T}$$

Convert $(V) \rightarrow (f)$

$f_i = f_c + A_m K_f$

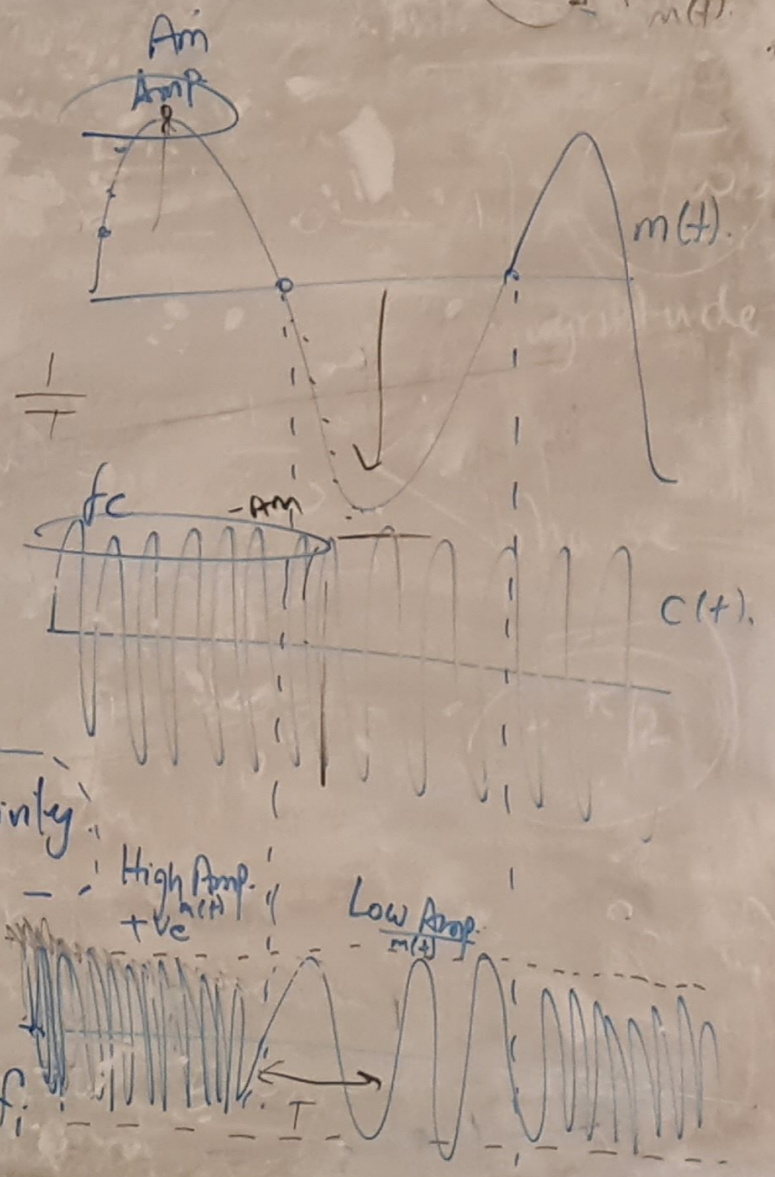
instantaneous freq

frequency sensitivity

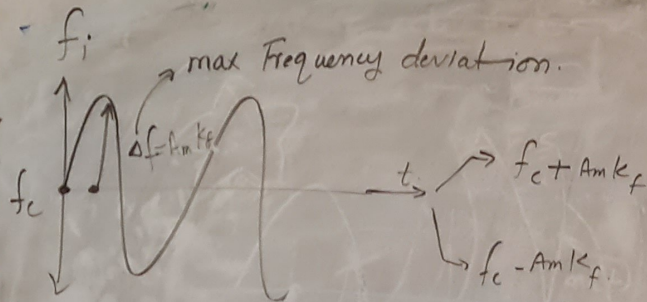
- at Rx.

Zero crossing detectors.

+ Freq to Voltage convert f_i



Angle Modulation FM / PM



Frequency Modulation.

$$\begin{aligned} S_{FM}(t) &= A_c \cos(\phi_i) \\ &= A_c \cos[2\pi \int (f_i) dt] \\ &= A_c \cos[2\pi \int [f_c + k_f m(t)] dt] \end{aligned}$$

$$S_{FM}(t) = A_c \cos(\omega_c t + \frac{k_f A_m}{f_m} \sin \omega_m t)$$

$$= A_c \cos(\omega_c t) + \beta \sin \omega_m t$$

$$\beta = \text{FM Modulation Index} = \frac{k_f A_m}{f_m} = \frac{\Delta f}{f_m}$$

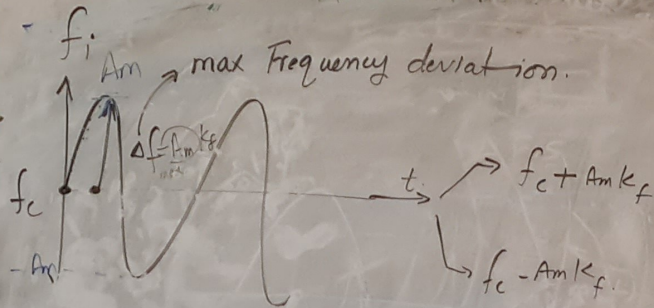
$$S(t) = A_c \cos(\phi_i)$$

in PM

$$\begin{aligned} S_{PM}(t) &= A_c \cos(\omega_c t + k_p m(t)) \\ &\quad \rightarrow \text{phase sensitivity factor} \end{aligned}$$

$$\beta = k_p A_m$$

$$\begin{aligned} S_{PM}(t) &= A_c \cos(2\pi f_c t + \beta \cos \omega_m t) \\ &\quad \rightarrow \text{Phase Mod. Index} \end{aligned}$$



Angle Modulation

FM / PM

$$S(t) = A_c \cos(\phi(t))$$

Frequency Modulation.
Expression.

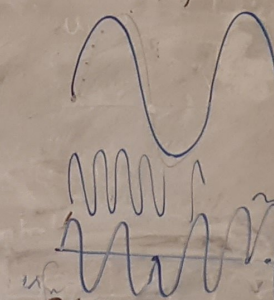
$$\begin{aligned} S_{FM}(t) &= A_c \cos(\phi(t)) \\ &= A_c \cos\left[2\pi \int (f_c + k_f m(t)) dt\right] \\ &= A_c \cos\left[2\pi \left[f_c t + k_f \int m(t) dt\right]\right] \end{aligned}$$

$$S_{FM}(t) = A_c \cos\left(\omega_c t + \frac{k_f A_m}{f_m} \sin \omega_m t\right)$$

$$S_{FM}(t) = A_c \cos\left[\omega_c t + \beta \sin \omega_m t\right]$$

$$P_t = \frac{A_c^2}{2R}$$

$$\beta = \text{FM Modulation Index} = \frac{k_f A_m}{f_m} = \frac{\Delta f}{f_m}$$

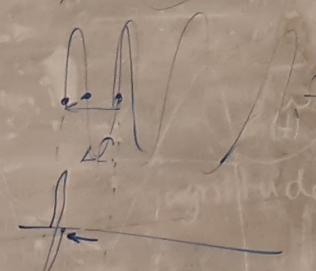


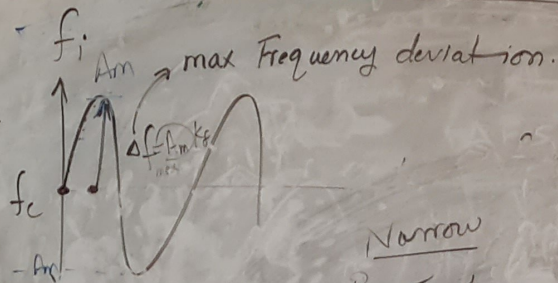
$$S_{PM}(t) = A_c \cos\left(2\pi f_c t + k_p m(t)\right)$$

$$\text{Ph. Mod. Index } \beta = k_p A_m$$

$$S_{PM}(t) = A_c \cos\left(2\pi f_c t + \beta \cos \omega_m t\right)$$

Phase Mod. Index



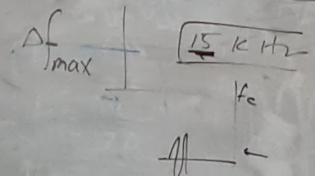
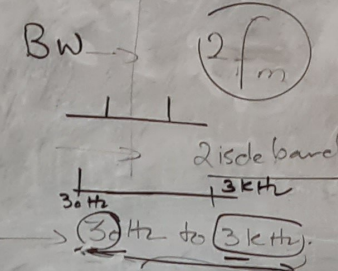


$20 \text{ kHz} \leftrightarrow 20 \text{ Hz}$

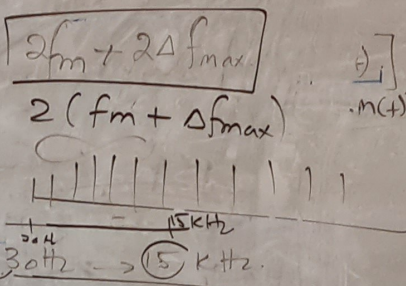
Angle Modulation

FM / PM

Narrow
 $\beta < 1$



wide
 $\beta > 1$



$$S(t) = A_c \cos(\omega_c t + \beta \sin(\omega_m t))$$

$$S(t) = 20 \cos(8\pi \times 10^6 t + 9 \sin(2\pi \times 10^3 t))$$

$\beta > 1 \rightarrow$ wide band

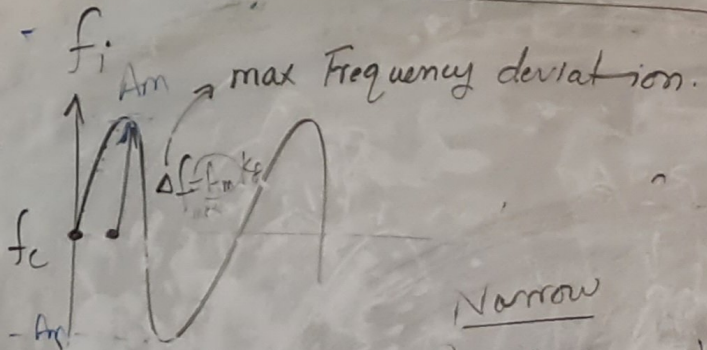
Find: Freq. deviation. - BW
Power.

$$\Delta f = k_f A_m$$

$$2(f_m + \Delta f_{max})$$

freq. deviation

20 kHz ← 20 Hz



Angle Modulation

FIM / PM

$$S(t) = A_c \cos(\dots)$$

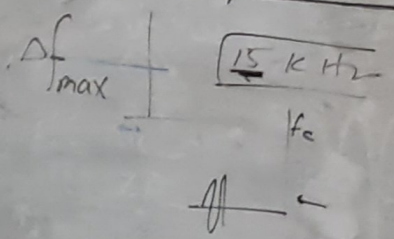
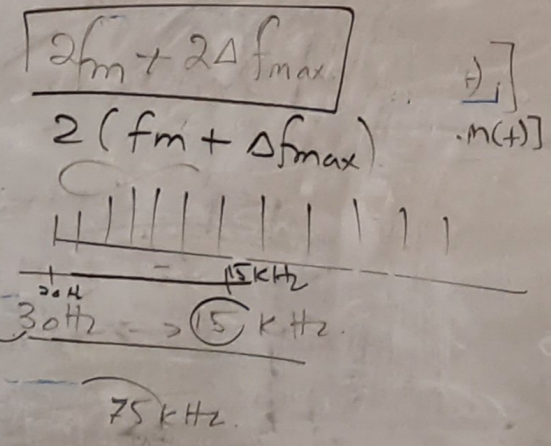
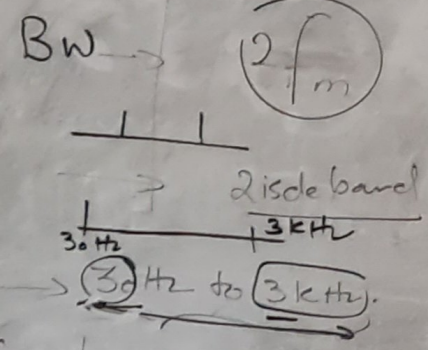
$$\frac{4\pi \times 10^6}{2\pi f_c} \Delta f$$

$$S(t) = 20 \cos(8\pi \times 10^6 t + 9 \dots)$$

$\beta > 1$

Narrow
 $\beta < 1$

Wide
 $\beta > 1$



Find. Freq. deviation
Power

$$P = \frac{A_c^2}{2R}$$

$$\Delta f = k_f A_m$$

$$\beta = \frac{\Delta f}{f_m}$$

$$\Delta f = \beta f_m$$

FM Modulation

FIM / PM

$$S(t) = A_c \cos(\omega_c t + \phi)$$

$\omega_c \uparrow$ m(H)

$$S(t) = 20 \cos(8\pi \times 10^6 t + 9 \sin(2\pi \times 10^3 t))$$

A_c $\frac{4 \times 10^6}{\omega_c f_c}$ β ω_m

$\beta > 1 \rightarrow$ wideba

Find.

Freq. deviation. — BW —

Power.

$$P = \frac{A_c^2}{2R}$$

$\frac{1}{2}$

$$\Delta f = (K_f) A_m$$

$$\beta = \frac{\Delta f}{f_m}$$

$$\Delta f = \beta f_m$$

10^3

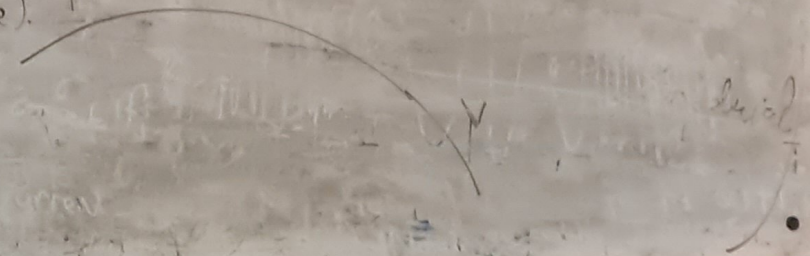
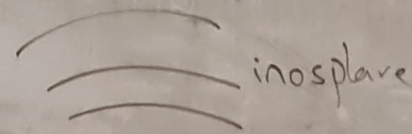
$$2(f_m + \Delta f_{max})$$

Freq. deviation

Angle Modulation

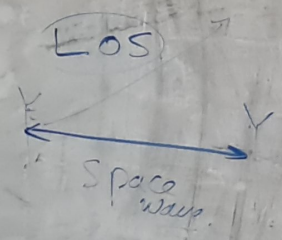
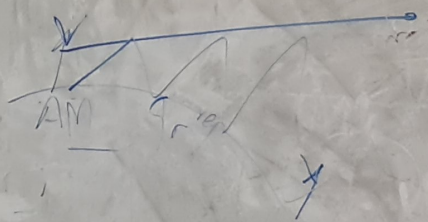
FIM / PM

AM	FM
HF - MF 3 - 30 MHz	VHF - UHF 30M - 300M 300M - 3G
distortion at atmosphere (storm)	Immune to space weather.
- propagation ground / sky wave - large distance	propagation (space wave)

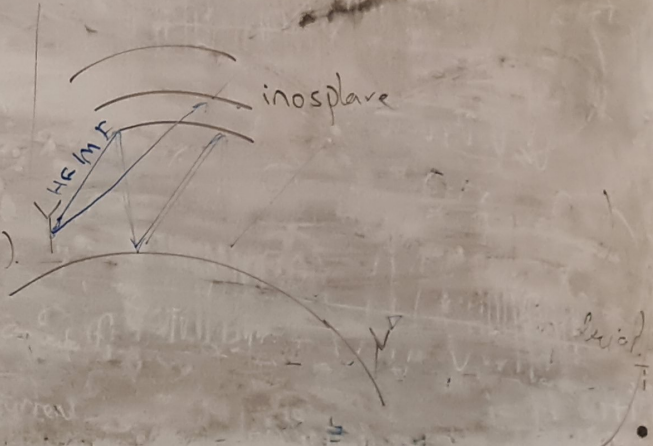


20 kHz $\longleftrightarrow$ 20 Hz

Angle Modulation FM / PM



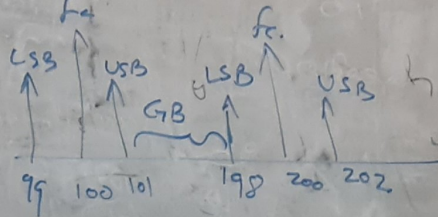
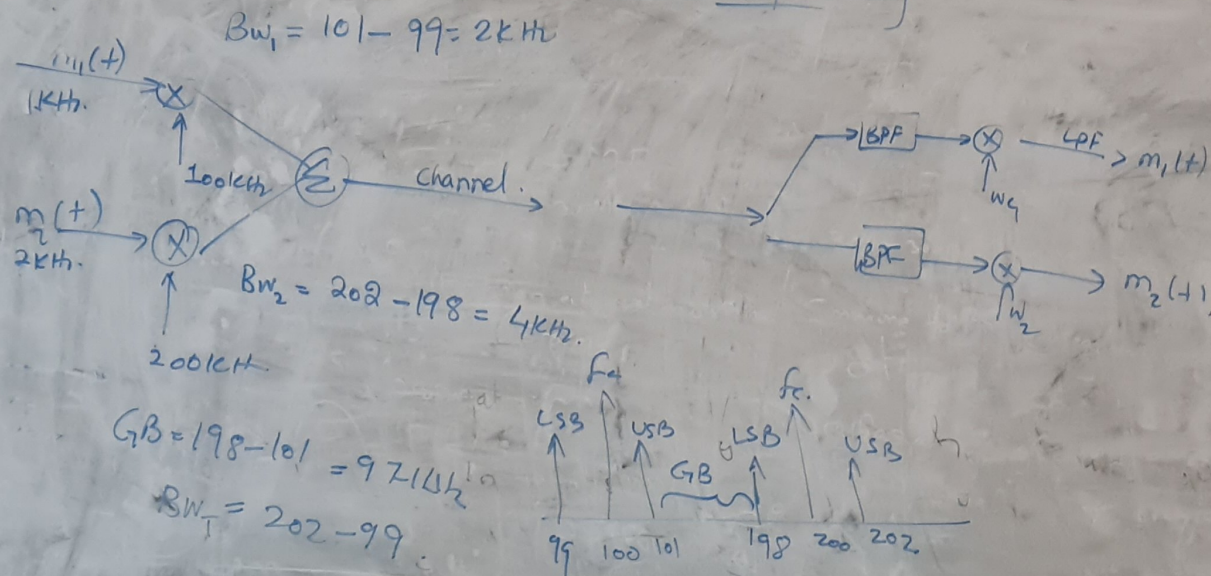
AM	FM
HF - MF 3 - 30 MHz	VHF - UHF 30M - 300M 300M - 3G
- distortion at atmosphere (storm) - propagation: ground / sky wave - large distance	- Immune to space weather - propagation: (Space wave)
LOS	LOS



FDM

Frequency division.

Multiplexing



FDM

Frequency division.

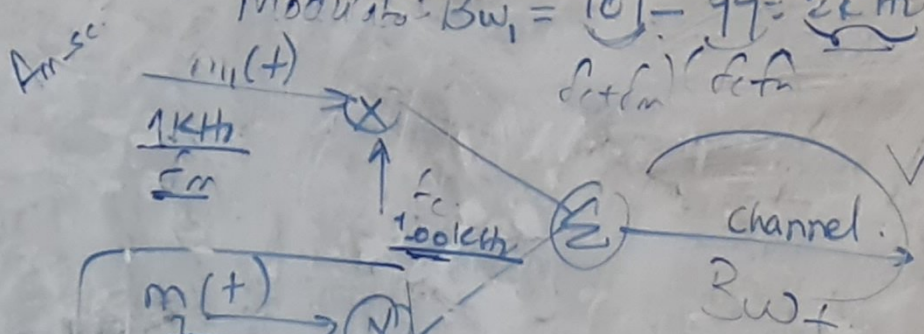
Multiplexing

$$Bw = 2f_m$$

$$= f_c + f_m - f_c - f_m$$

Modulator: $Bw_1 = 101 - 99 = 2 \text{ kHz}$

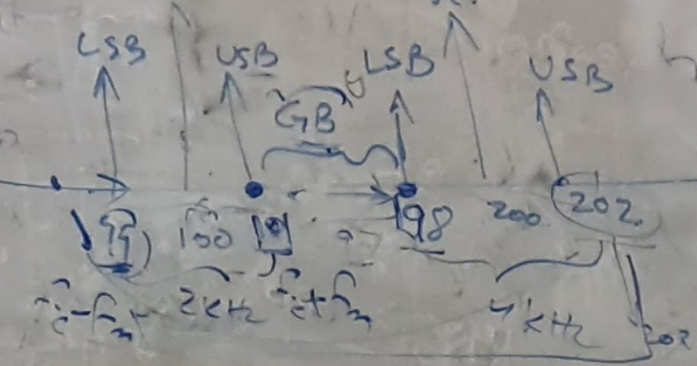
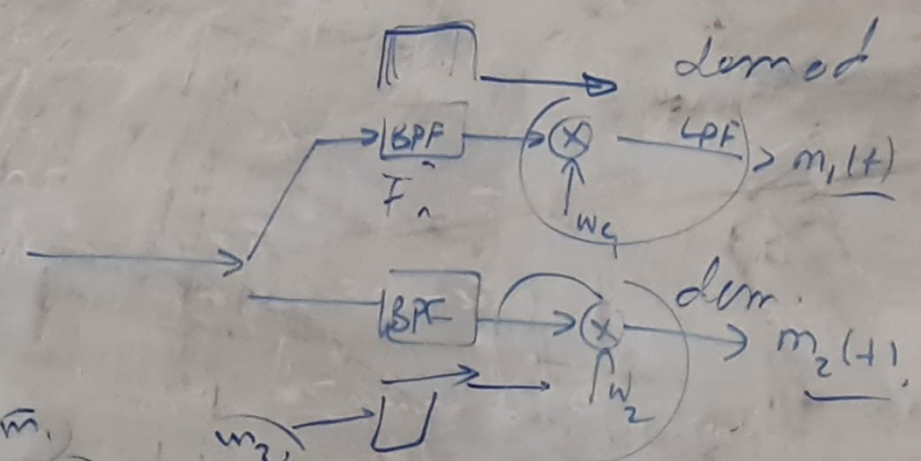
$$f_c + f_m$$



$$Bw_2 = \frac{202}{2} - 198 = 4 \text{ kHz}$$

$$GB = 198 - 191 = 9 \text{ kHz}$$

$$Bw_T = 202 - 99$$



Frequency division Multiplexing

Signals are Transmitted through a channel.

$$\rightarrow 10 \cos 500t + 15 \cos 800t + 12 \cos 1000t$$

are modulated with the following carrier. respectively.

20 kHz, 30 kHz, 50 kHz.

Find:

1. Signals Bandwidth

2. Total BW

3. Guardband (GB) between signals

4. draw the Frequency spectrum